Investigating the Relationship Between Students' Lifestyle Choices and Academic Performance

Empirical Analysis, Monte Carlo Sampling Verification, and Multivariate Lifestyle Modeling

Quantitative Research Report

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Abstract

This report investigates the empirical relationships among daily time allocation, self-reported psychological stress, and cumulative grade point average (CGPA) across a cohort of $N = 2,000$ students. Using descriptive profiling, stress-stratified distributional comparisons, Pearson correlation analysis, Monte Carlo Central Limit Theorem (CLT) simulations, and ordinary least squares (OLS) regression modeling, we evaluate how five core lifestyle dimensions—study hours, sleep duration, extracurricular involvement, social interaction, and physical activity—interact with academic achievement. The empirical findings demonstrate that daily study duration exhibits a strong positive linear correlation with CGPA ($r = 0.734$), whereas physical activity exhibits a moderate negative correlation ($r = -0.341$) largely attributable to the fixed 24-hour daily time-budget constraint. Notably, stratification by stress level reveals an achievement–stress trade-off: students in the high-stress tier (51.45% of the cohort) achieve the highest mean CGPA (3.26) and log the longest study hours (8.39 hours/day) while sleeping less (7.05 hours/day) than peers in moderate- and low-stress tiers. Finally, multivariate linear models explain 54.10% of the variance in CGPA ($R^2 = 0.5410$) and 64.77% of the variance in encoded stress levels ($R^2 = 0.6477$), motivating a constrained linear optimization framework for balancing academic targets (CGPA ≥ 3.0) against health and well-being constraints (Sleep ≥ 6.0 hours/day, Stress ≤ 1).

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1 Introduction and Dataset Overview

Understanding how students allocate their limited daily hours across academic, restorative, physical, and social activities is central to both educational psychology and institutional student-support policy. While intensive study habits are conventionally associated with higher academic standing, excessive academic time commitments can crowd out sleep and physical exercise, elevating psychological stress. Conversely, insufficient study time may preserve leisure and lower acute stress at the expense of academic performance.

1.1 Research Objectives

The primary objective of this study is to quantify the empirical links between students' daily lifestyle habits, stress levels, and academic performance using the student lifestyle dataset (`student_lifestyle_dataset.csv`). Specifically, the analysis is structured around four core inquiries:

1. **Distributional Profiling:** Characterizing the central tendency, dispersion, and quartile structure of daily routine variables and academic outcomes across the student population.
2. **Stress Stratification and Correlation Analysis:** Evaluating how academic performance and time-use habits vary across Low, Moderate, and High stress groups, and quantifying bivariate linear associations with CGPA.
3. **Asymptotic Sampling Verification:** Conducting a Monte Carlo simulation on daily study hours to empirically verify the Central Limit Theorem (CLT) across increasing sample sizes $n \in \{1, 2, 5, 10, 30, 100\}$.
4. **Predictive Regression and Constrained Lifestyle Optimization:** Fitting bivariate and multivariate linear regression models for CGPA and stress to establish a quantitative decision framework that satisfies academic targets without inducing severe sleep deprivation or high stress.

1.2 Data Integrity and Schema

The dataset comprises exactly $N = 2,000$ student records across 8 attributes. Initial data quality auditing confirmed complete data integrity: there are 0 missing values (`NaN`) and 0 duplicate rows across the entire dataset. Table 1 displays the first five records of the dataset, illustrating the raw feature representation.

The eight variables in the schema are defined as follows:

- `Student_ID`: Unique integer identifier ranging from 1 to $2,000$ (excluded from statistical modeling).
- `Study_Hours_Per_Day`: Average daily hours spent on academic study outside class.
- `Extracurricular_Hours_Per_Day`: Average daily hours dedicated to clubs, organizations, or structured co-curricular activities.
- `Sleep_Hours_Per_Day`: Average daily hours of sleep.
- `Social_Hours_Per_Day`: Average daily hours spent on social activities and peer interaction.

Table 1: Sample Student Records from `student_lifestyle_dataset.csv` (First 5 Observations)

Student ID	Study (h)	Extracurr. (h)	Sleep (h)	Social (h)	Physical (h)	CGPA	Stress Level
1	6.9	3.8	8.7	2.8	1.8	2.99	1 (Moderate)
2	5.3	3.5	8.0	4.2	3.0	2.75	0 (Low)
3	5.1	3.9	9.2	1.2	4.6	2.67	0 (Low)
4	6.5	2.1	7.2	1.7	6.5	2.88	1 (Moderate)
5	8.1	0.6	6.5	2.2	6.6	3.51	2 (High)

- `Physical_Activity_Hours_Per_Day`: Average daily hours spent on exercise, sports, or physical recreation.
- `CGPA` (also referenced as `GPA` in modeling modules): Cumulative Grade Point Average on a standard 4.00 scale.
- `Stress_Level`: Categorical assessment of perceived student stress, represented either as ordered labels (`Low`, `Moderate`, `High`) or ordinally encoded integers (0 = Low, 1 = Moderate, 2 = High).

1.3 Compositional 24-Hour Time-Budget Structure

A critical structural property of the five daily time-use features is that they partition a complete 24-hour day. For any individual student $i \in \{1, \ldots, 2000\}$, let $H_{i,\text{study}}$, $H_{i,\text{extra}}$, $H_{i,\text{sleep}}$, $H_{i,\text{social}}$, and $H_{i,\text{phys}}$ denote the five reported daily activity durations. Inspection of the records in Table 1 shows that for each student:

$$H_{i,\text{study}} + H_{i,\text{extra}} + H_{i,\text{sleep}} + H_{i,\text{social}} + H_{i,\text{phys}} = 24.0 \text{ hours} \tag{1}$$

For example, for Student 1, $6.9 + 3.8 + 8.7 + 2.8 + 1.8 = 24.0$ hours. Recognizing Equation 1 is essential for interpreting bivariate correlations and multivariate regression models: because total daily time is conserved at 24.0 hours, increasing time in one domain (such as study) necessarily requires a compensating reduction across one or more of the remaining four domains.

2 Exploratory Data Analysis of Daily Routines and CGPA

To establish a quantitative baseline across the $N = 2,000$ students, summary statistics (mean, standard deviation, minimum, 25th percentile, median, 75th percentile, and maximum) were computed for all seven analytical variables after dropping `Student_ID`. Table 2 presents the complete distributional summary.

Table 2: Descriptive Statistics of Student Lifestyle Variables and Academic Performance ($N = 2,000$)

Variable	Count	Mean	SD	Min	25%	50%	75%	Max
Study Hours / Day	2,000	7.48	1.42	5.00	6.30	7.40	8.70	10.00
Extracurricular Hours / Day	2,000	1.99	1.16	0.00	1.00	2.00	3.00	4.00
Sleep Hours / Day	2,000	7.50	1.46	5.00	6.20	7.50	8.80	10.00
Social Hours / Day	2,000	2.70	1.69	0.00	1.20	2.60	4.10	6.00
Physical Activity Hours / Day	2,000	4.33	2.51	0.00	2.40	4.10	6.10	13.00
CGPA (4.00 Scale)	2,000	3.12	0.30	2.24	2.90	3.11	3.33	4.00
Stress Level (0–2 Scale)	2,000	1.37	0.73	0.00	1.00	2.00	2.00	2.00

2.1 Distributional Characteristics of Daily Routines

Several empirical patterns emerge from the summary statistics in Table 2 and the corresponding 15-bin kernel density histograms generated during exploratory analysis:

- **Study and Sleep Bounded Symmetry:** Both `Study_Hours_Per_Day` ($\mu = 7.48$, SD $= 1.42$, median $= 7.40$) and `Sleep_Hours_Per_Day` ($\mu = 7.50$, SD $= 1.46$, median $= 7.50$) span the exact bounded interval $[5.00, 10.00]$ hours/day. Their means and medians are nearly identical, and their interquartile ranges ($\text{IQR}_{\text{study}} = 2.40$ hours, $\text{IQR}_{\text{sleep}} = 2.60$ hours) reflect broad, near-uniform dispersion across the $[5, 10]$ interval rather than a sharply peaked Gaussian profile.
- **Extracurricular and Social Commitments:** `Extracurricular_Hours_Per_Day` is symmetrically distributed across $[0.00, 4.00]$ hours/day with a mean of 1.99 hours (SD $= 1.16$) and exact integer quartiles (1.00, 2.00, and 3.00 hours/day). `Social_Hours_Per_Day` spans $[0.00, 6.00]$ hours/day with a mean of 2.70 hours (SD $= 1.69$) and a median of 2.60 hours.
- **Physical Activity Dispersion:** `Physical_Activity_Hours_Per_Day` exhibits the widest absolute range ($[0.00, 13.00]$ hours/day) and largest standard deviation (SD $= 2.51$ hours) among all routine variables, with a mean of 4.33 hours and median of 4.10 hours (indicating slight right-skewness). Consistent with Equation 1, the sum of the five sample means equals $7.48 + 1.99 + 7.50 + 2.70 + 4.33 = 24.00$ hours.
- **Academic Performance (CGPA):** Student grades follow a well-behaved, unimodal, approximately bell-shaped distribution centered at a mean of 3.12 (SD $= 0.30$) and a median of 3.11. The central 50% of students earn CGPAs between 2.90 and 3.33, with observed scores spanning 2.24 to 4.00.

3 Stress Level Stratification and Correlation Analysis

3.1 Prevalence of Student Stress

Evaluating the distribution of students across self-reported stress categories reveals that elevated stress is the predominant condition in this cohort. Table 3 summarizes the absolute counts, percentages, and group-level lifestyle averages across the three stress tiers.

As shown in Table 3, more than half of all surveyed students ($1,029$ students, or 51.45%) fall into the **High** stress category. Another 674 students (33.70%) report **Moderate** stress, leaving only

Table 3: Cohort Distribution and Mean Lifestyle Profiles Stratified by Stress Level ($N = 2,000$)

Stress Tier	Encoded	Students (n)	Share (%)	Mean CGPA	Mean Study (h)	Mean Sleep (h)
Low Stress	0	297	14.85%	2.82	5.47	8.06
Moderate Stress	1	674	33.70%	3.02	6.97	7.95
High Stress	2	1,029	51.45%	3.26	8.39	7.05
Total / Overall	**1.37**	**2,000**	**100.00%**	**3.12**	**7.48**	**7.50**

297 students (14.85%) in the **Low** stress category. This skew toward high stress explains why the encoded `Stress_Level` variable in Table 2 has a mean of 1.37 and a median (50th percentile) equal to 2.00.

3.2 Group Averages and Routine Comparisons Across Stress Tiers

To examine how daily routines and academic outcomes co-vary with stress, box plots and group means were computed across the three stress groups (Table 4).

Table 4: Comparison of Group Averages Across Stress Levels

Stress Level	Mean CGPA	Study Hours/Day	Sleep Hours/Day	Physical Activity Hours/Day
Low	2.82	5.47	8.06	5.58
Moderate	3.02	6.97	7.95	4.34
High	3.26	8.39	7.05	3.96
Net Shift (High − Low)	$+0.44$	$+2.92$ **h**	-1.01 **h**	-1.62 **h**

Comparing the three stress tiers highlights a striking behavioral gradient:

1. **Monotonic Rise in Study Hours and CGPA:** Students in the Low stress group study an average of 5.47 hours/day and achieve a mean CGPA of 2.82. In the Moderate stress group, average study time rises by 1.50 hours/day to 6.97 hours/day, accompanied by a $+0.20$ increase in mean CGPA (3.02). In the High stress group, average study duration reaches 8.39 hours/day ($+2.92$ hours/day above the Low stress group) and mean CGPA reaches 3.26 ($+0.44$ above the Low stress group).

2. **Sleep Compression in High Stress:** Average sleep duration remains high and nearly identical between Low stress (8.06 hours/day) and Moderate stress (7.95 hours/day), but drops sharply by nearly a full hour to 7.05 hours/day in the High stress group as study demands expand and sleep falls below restorative thresholds for many students.

3. **Decline in Physical Activity:** Mean physical activity decreases progressively from 5.58 hours per day in the Low stress tier to 4.34 hours/day in the Moderate tier and 3.96 hours/day in the High stress tier ($\Delta = -1.62$ hours/day). By contrast, box plots of social hours and extracurricular hours show substantially overlapping interquartile ranges across all three stress levels.

3.3 Bivariate Correlations with CGPA

To quantify the linear association between each daily routine variable and academic performance across the full cohort ($N = 2,000$), Pearson product-moment correlation coefficients (r) were calculated against `CGPA`:

$$r_{X,Y} = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{N}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{N}(y_i - \bar{y})^2}} \tag{2}$$

Table 5 ranks the five lifestyle variables by their Pearson correlation with CGPA.

Table 5: Pearson Correlation Coefficients (r) Between Daily Routine Variables and CGPA ($N = 2,000$)

Lifestyle Variable	**Correlation (r)**	**Empirical Interpretation**
`Study_Hours_Per_Day`	0.734	Strong positive linear relationship
`Sleep_Hours_Per_Day`	-0.004	Negligible / near-zero linear relationship
`Extracurricular_Hours_Per_Day`	-0.032	Negligible negative relationship
`Social_Hours_Per_Day`	-0.086	Very weak negative relationship
`Physical_Activity_Hours_Per_Day`	-0.341	Moderate negative linear relationship

Key Analytical Finding. In this dataset, daily study hours are the dominant positive correlate of CGPA ($r = 0.734$, accounting for $r^2 \approx 53.9\%$ of variance alone). Sleep hours ($r = -0.004$), extracurricular hours ($r = -0.032$), and social hours ($r = -0.086$) exhibit virtually zero marginal linear correlation with CGPA. Physical activity exhibits a moderate negative correlation ($r = -0.341$) because, under the 24-hour compositional budget, high-studying students allocate significantly fewer residual hours to physical activity.

It is vital to emphasize that these associations are observational and descriptive of this specific cohort; they do not establish that higher stress or reduced physical activity causally improves academic performance. Rather, both elevated CGPA and elevated stress are joint consequences of intensive study schedules coupled with reduced sleep.

4 Sampling Theory Verification: Central Limit Theorem Simulation

Beyond the primary empirical analysis of academic performance, a Monte Carlo simulation was conducted on the dataset to verify the **Central Limit Theorem (CLT)** using a real behavioral variable whose underlying population distribution departs substantially from normality.

4.1 Theoretical Framework and Simulation Design

As noted in Section 2, `Study_Hours_Per_Day` is bounded on $[5.0, 10.0]$ with a near-uniform, platykurtic shape rather than a normal bell curve. Treating the $N = 2,000$ observations of `Study_Hours_Per_Day` as a finite empirical population yields exact population parameters (using population divisor N, i.e., ddof $= 0$):

$$\mu = 7.4758 \text{ hours/day}, \qquad \sigma = 1.4235 \text{ hours/day} \tag{3}$$

By the Central Limit Theorem, when independent random samples of size n are drawn with replacement from any population with finite mean μ and finite variance σ^2, the sampling distribution of the sample mean $\bar{X}_n = \frac{1}{n}\sum_{j=1}^{n} X_j$ converges in distribution to a Gaussian random variable as $n \to \infty$:

$$\bar{X}_n \xrightarrow{d} \mathcal{N}\left(\mu,\ \frac{\sigma^2}{n}\right), \qquad \text{with theoretical standard error } \mathrm{SE}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}} \tag{4}$$

To test Equation 4 empirically, a reproducible pseudo-random number generator seeded at zero (`np.random.default_rng(0)`) was used to draw $M = 5,000$ independent bootstrap samples with replacement for each of six sample sizes:

$$n \in \{1,\ 2,\ 5,\ 10,\ 30,\ 100\}$$

For each sample size n, the empirical mean of the $5,000$ sample means, the empirical sample standard deviation (ddof $= 1$), the theoretical CLT standard error ($\sigma/\sqrt{n}$), and the third standardized moment (Fisher-Pearson skewness) were computed.

4.2 Simulation Results and Convergence Analysis

Table 6 reports the complete numerical results of the Monte Carlo simulation across all six sample sizes.

Table 6: Monte Carlo Central Limit Theorem Simulation on `Study_Hours_Per_Day` ($5,000$ Draws per n; Population $\mu = 7.4758$, $\sigma = 1.4235$)

Sample Size (n)	**Mean of $\bar{X}_n$**	**Empirical SD of $\bar{X}_n$**	**CLT SD ($\sigma/\sqrt{n}$)**	**Absolute Error**	**Skewness**
$n = 1$	7.4766	1.4223	1.4235	0.0012	0.0323
$n = 2$	7.4637	1.0017	1.0066	0.0049	0.0383
$n = 5$	7.4583	0.6477	0.6366	0.0111	0.0769
$n = 10$	7.4798	0.4554	0.4502	0.0052	0.0047
$n = 30$	7.4793	0.2585	0.2599	0.0014	−0.0345
$n = 100$	7.4763	0.1437	0.1424	0.0013	0.0691

The simulation results in Table 6 provide textbook confirmation of asymptotic sampling theory:

1. **Unbiasedness of the Sample Mean:** Across all six values of n, the empirical average of the $5,000$ sample means remains tightly anchored to the true population mean $\mu = 7.4758$, fluctuating within a narrow band of $[7.4583, 7.4798]$ (a maximum deviation of less than 0.24%).

2. **Inverse Square-Root Shrinkage of Dispersion:** As n increases from 1 to 100 (a 100-fold increase in sample size), the empirical standard error shrinks by an exact factor of $\sqrt{100} = 10$, dropping from 1.4223 at $n = 1$ to 0.1437 at $n = 100$ (matching the theoretical prediction $\sigma/\sqrt{100} = 0.1424$ to within 0.0013 hours).

3. **Rapid Normal Approximation:** At $n = 1$, the density histogram reproduces the flat, bounded population distribution on $[5, 10]$. At $n = 2$, the convolution of two uniform-like distributions produces a triangular density profile. By $n = 5$ and $n = 10$, the empirical density aligns closely with the overlaid dashed Gaussian curve $\mathcal{N}(\mu, \sigma^2/n)$, and for $n \geq 30$ the sampling distribution is visually indistinguishable from a normal distribution with near-zero skewness ($|\gamma_1| < 0.08$).

5 Predictive Modeling and Lifestyle Optimization Framework

To move from descriptive correlation to predictive inference and decision support, ordinary least squares (OLS) linear regression models were estimated using `scikit-learn`.

5.1 Bivariate Regression Model: Study and Sleep Hours on GPA

First, a two-predictor linear regression model was fitted using `Study_Hours_Per_Day` (X_{study}) and `Sleep_Hours_Per_Day` (X_{sleep}) to predict `GPA` (Y). The estimated OLS regression hyperplane is given by:

$$\widehat{\text{GPA}} = \beta_0 + \beta_{\text{study}} X_{\text{study}} + \beta_{\text{sleep}} X_{\text{sleep}} \tag{5}$$

Fitting the model across all $N = 2,000$ students yields the exact parameter estimates reported in Table 7:

$$\widehat{\text{GPA}} = 1.999906 + 0.154195\, X_{\text{study}} - 0.004890\, X_{\text{sleep}} \tag{6}$$

Table 7: Parameter Estimates for the Bivariate Linear Regression Model of GPA ($N = 2,000$)

Parameter	**Variable**	**Estimate**	**Marginal Interpretation**
Intercept (β_0)	Constant	1.999906	Baseline predicted GPA at zero study and sleep hours
Slope (β_{study})	`Study_Hours_Per_Day`	0.154195	$+0.1542$ GPA points per additional study hour/day
Slope (β_{sleep})	`Sleep_Hours_Per_Day`	-0.004890	-0.0049 GPA points per additional sleep hour/day

Visualizing Equation 6 as a 3D response surface over a 20×20 meshgrid of study hours ($[5.0, 10.0]$) and sleep hours ($[5.0, 10.0]$), alongside its 2D filled contour projection, demonstrates that the GPA gradient is almost entirely parallel to the study-hours axis. Holding sleep constant, each additional hour of daily study increases expected GPA by approximately 0.154 points (meaning a 5-hour increase from 5 to 10 hours/day raises predicted GPA by $+0.771$ points, from ≈ 2.73 to ≈ 3.50). Conversely, holding study time constant, varying sleep across its entire 5-hour range alters predicted GPA by only $5 \times (-0.004890) = -0.024$ points, rendering the contour lines nearly vertical in the $(\text{Study}, \text{Sleep})$ plane.

5.2 Multivariate Regression Models for GPA and Stress Level

Next, full-feature linear regression models were fitted using the lifestyle feature matrix $\mathbf{X}$ to predict both continuous academic performance (`gpa_model`) and ordinally encoded psychological stress (`stress_model`, where Low $= 0$, Moderate $= 1$, High $= 2$):

$$\widehat{\text{GPA}}(\mathbf{x}) = b_0 + \mathbf{b}^\top \mathbf{x} \tag{7}$$

$$\widehat{\text{Stress}}(\mathbf{x}) = s_0 + \mathbf{s}^\top \mathbf{x} \tag{8}$$

Table 8 summarizes the goodness-of-fit (R^2, coefficient of determination) achieved by both multivariate models across the $N = 2,000$ student cohort.

Table 8: Goodness-of-Fit (R^2) for Full-Feature Multivariate Linear Models ($N = 2,000$)

Model Name	Target Variable	Exact R^2	Variance Explained	Model Fit Assessment
`gpa_model`	`GPA` $(2.24 - 4.00)$	0.540968	54.10%	Moderate-to-strong linear fit
`stress_model`	`Stress_Level` $(0, 1, 2)$	0.647705	64.77%	Strong linear explanatory power

Two important analytical insights follow from Table 8:

1. **Dominance of Study Hours in GPA Prediction:** Comparing the bivariate correlation squared of study hours alone ($r^2 = 0.734^2 = 0.5388$, or 53.88%) with the full multivariate model's $R^2 = 0.5410$ (54.10%) reveals that adding sleep, extracurricular, social, and physical activity hours increases explained GPA variance by only $+0.22$ percentage points. Thus, within this dataset, linear predictability of GPA is overwhelmingly driven by study duration.

2. **High Linear Predictability of Stress:** The multivariate stress model explains 64.77% of the variance in encoded stress levels ($R^2 = 0.6477$). Unlike GPA—which depends almost solely on study hours—student stress is jointly sensitive to both high study hours (which elevate stress) and low sleep hours (which sharply exacerbate stress when sleep falls below 6–7 hours/day).

5.3 Constrained Lifestyle Optimization Formulation

Using the extracted regression parameter vectors $(\mathbf{b}, b_0)$ from `gpa_model` and $(\mathbf{s}, s_0)$ from `stress_model`, the notebook establishes a quantitative target specification for student lifestyle design:

- **Minimum Academic Performance Target:** target_gpa $= 3.00$ (maintaining at least a B average).
- **Minimum Restorative Sleep Threshold:** min_sleep $= 6.00$ hours/day, explicitly chosen to lie approximately one standard deviation (SD $= 1.46$ hours) below the cohort mean sleep duration ($\mu_{\text{sleep}} = 7.50$ hours/day).
- **Maximum Acceptable Stress Ceiling:** max_stress $= 1.00$ (restricting predicted stress to at most the `Moderate` tier and avoiding the `High` stress regime).

Combining Equations 1, 7, and 8 with these thresholds yields a formal linear feasibility and optimization system over the daily activity vector $\mathbf{x} = (x_{\text{study}}, x_{\text{extra}}, x_{\text{sleep}}, x_{\text{social}}, x_{\text{phys}})^\top$:

$$\begin{aligned}
\text{Find / Optimize} \quad & \mathbf{x} \in \mathbb{R}^5_+ \\
\text{subject to} \quad & b_0 + \mathbf{b}^\top \mathbf{x} \geq 3.00 && \text{(Target GPA: target_gpa} = 3.0\text{)} \\
& s_0 + \mathbf{s}^\top \mathbf{x} \leq 1.00 && \text{(Stress ceiling: max_stress} = 1.0\text{)} \\
& x_{\text{sleep}} \geq 6.00 && \text{(Sleep floor: } \mu_{\text{sleep}} - 1\sigma \approx 6.0 \text{ h)} \\
& \sum_{k=1}^{5} x_k = 24.00 && \text{(24-hour time-budget conservation)} \\
& l_k \leq x_k \leq u_k && \text{(Empirical bounds from Table 2)}
\end{aligned} \tag{9}$$

Evaluating the bivariate benchmark (Equation 6) at the sleep floor $x_{\text{sleep}} = 6.0$ hours/day shows that achieving $\widehat{\text{GPA}} \geq 3.00$ requires:

$$\begin{aligned}
1.999906 + 0.154195\, x_{\text{study}} - 0.004890(6.0) &\geq 3.00 \\
0.154195\, x_{\text{study}} \geq 1.029434 \implies x_{\text{study}} &\geq 6.68 \text{ hours/day}
\end{aligned}$$

If a student instead maintains the Moderate-stress group's average sleep duration ($x_{\text{sleep}} = 7.95$ hours/day), achieving $\widehat{\text{GPA}} = 3.00$ requires $x_{\text{study}} \approx 6.74$ hours/day—which closely aligns with the empirical Moderate-stress group profile observed in Table 4 (Study $= 6.97$ h/day, Sleep $= 7.95$ h/day, CGPA $= 3.02$, Stress $= 1$).

Key Analytical Finding. The Moderate-stress empirical regime (≈ 6.7–7.0 hours/day of study and ≈ 7.5–8.0 hours/day of sleep) represents the practical "sweet spot" in this dataset: it reliably clears the CGPA ≥ 3.00 threshold while preserving restorative sleep well above the 6.0-hour minimum and avoiding the transition into the High-stress tier.

6 Methodological Limitations and Conclusion

6.1 Scope and Methodological Limitations

When interpreting the statistical findings of this report, three methodological constraints should be kept in mind:

1. **Observational Cross-Sectional Design:** All metrics in `student_lifestyle_dataset.csv` represent observational, cross-sectional records rather than longitudinal or randomized interventions. Consequently, high correlations (such as $r = 0.734$ between study hours and CGPA, or higher mean CGPA among high-stress students) reflect statistical association rather than direct causation.

2. **Perfect Multicollinearity of the 24-Hour Time Budget:** Because the five daily routine variables sum identically to 24.0 hours for every observation ($\sum_{k=1}^{5} x_{ik} = 24$), including all five routine variables simultaneously alongside an unconstrained intercept in an ordinary least squares regression introduces exact structural collinearity ($\sum_{k=1}^{5} x_{ik} = 24 \cdot 1$). While prediction scores ($R^2 = 0.5410$ and $R^2 = 0.6477$) remain invariant to this collinearity, individual partial regression slopes in the 5-feature model must be interpreted relative to the 24-hour simplex constraint.

3. **Linear Approximation of Ordinal Stress:** Encoding `Stress_Level` onto the integer scale $\{0, 1, 2\}$ treats the psychological distance between Low and Moderate stress as identical to the distance between Moderate and High stress. Future extensions could complement OLS with ordered probit/logit classification or multinomial logistic regression.

6.2 Summary of Conclusions

This investigation of $N = 2,000$ student lifestyle records yields four principal conclusions:

- **Study Duration Drives Academic Standing:** Daily study time is the primary positive predictor of CGPA ($r = 0.734$, bivariate slope $+0.1542$ GPA points/hour), accounting for nearly all explainable linear variance in student grades ($R^2 = 0.5410$).
- **High Achievement Often Co-Occurs with High Stress and Sleep Loss:** Over half the cohort (51.45%) experiences high stress. While high-stress students achieve the highest average CGPA (3.26), they do so by studying 8.39 hours/day and compressing sleep to 7.05 hours/day.
- **Empirical Validation of the Central Limit Theorem:** Monte Carlo bootstrap sampling ($5,000$ replications per sample size) on non-normal study hours ($\mu = 7.4758, \sigma = 1.4235$) confirms that sample means converge rapidly to $\mathcal{N}(\mu, \sigma^2/n)$, with empirical standard errors matching theoretical $\sigma/\sqrt{n}$ predictions to within 0.011 hours across all $n \in \{1, 2, 5, 10, 30, 100\}$.
- **Feasible Lifestyle Balancing:** Regression-based constraint analysis demonstrates that students can achieve a target CGPA ≥ 3.00 without entering the High-stress regime (Stress ≤ 1) by allocating approximately 6.7–7.0 hours/day to focused study while safeguarding at least 6.0–8.0 hours/day of sleep.